



Agency models in different stages of CEO tenure: The effects of stock options and board independence on R&D investment



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ABSTRACT

This study examines how R&D investment is shaped by governance devices. Combining the two agency models of limited competence and managerial opportunism, it suggests that governance devices exert differing effects, in early vs. later stages of CEO tenure. Early in CEO tenure, R&D investment is reduced by stock options and board independence, whereas in later stages these effects reverse: R&D investment is enhanced by stock options and board independence. An empirical test on a sample of U.S. companies provides support for the hypothesized effects. For policy makers, this study implies that R&D investment can be enhanced by setting rules and standards of good governance that take into account CEO time in office and firm context.

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1. Introduction

Decisions on R&D investment are at the core of corporate success. R&D investment fosters new technological processes and firm growth (Del Monte and Papagni, 2003; Geroski and Machin, 2006; Hay and Kamshad, 1994), enhances firm productivity (Harhoff, 1998; Kafourous, 2005; Tsai and Wang, 2004; Verspagen, 1995), and contributes to competitive advantage and superior corporate performance (Garcia-Manjon and Romero-Merino, 2012; Wang and Thornhill, 2010). At the country level, R&D investment contributes to GDP growth (Corrado et al., 2009; Driver and Guedes, 2012; Wakelin, 2001). Despite these benefits, a number of factors make executives reluctant to invest in R&D: R&D investment is risky (Lee and O'Neill, 2003), entails ambiguous cause-effect relations (Munari et al., 2010), and reduces performance in the near term, while postponing payoffs (if any) in the long term (Driver and Guedes, 2012). Hence, R&D investment has constantly been at the center of debate among scholars, policy makers, and practitioners, in an attempt to identify the determinants of R&D spending and prescriptions to foster innovation effort. In their extensive literature review, Ahuja et al. (2008) group the determinants of R&D investment into four broad headings: industry structure, firm characteristics, intra-organizational attributes, and institutional influences. Among intra-organizational attributes, corporate

governance represents a major determinant: governance mechanisms, such as stock options and board independence, influence the proclivity of managers to take risk, and consequently their incentives to invest in R&D (Ahuja et al., 2008).

This study contends that strong governance devices are not always beneficial for R&D investment, but their effects change, and may be even detrimental, at different stages of CEO tenure. Extant research recognizes that CEOs' psychological attitudes and power evolve over time, so that CEOs exhibit a different proclivity to invest, or not invest, in R&D, at differing stages of tenure (Ahuja et al., 2008; Hambrick and Fukutomi, 1991). It can thus be expected that stock options and boards of directors will have differing effects on R&D investment along the CEO tenure cycle. This inquiry is important for both theoretical and practical reasons.

Previous research on innovation has sought to ascertain whether R&D spending can be influenced by governance devices, but empirical evidence has been inconclusive (Baysinger et al., 1991; Hill and Snell, 1988; Driver and Guedes, 2012; Hoskisson et al., 2002). Most of these studies rely on the agency model of managerial opportunism, to suggest that CEOs tend to underinvest in R&D, so that board independence and stock options are needed to boost innovation effort. Yet, an agency problem of limited competence may also arise in principal-agent relations (Hendry, 2002), and in such instances governance devices may exert very differing effects on R&D investment. Combining both agency models may help explain the varied impacts of corporate governance in differing stages of CEO tenure, and offer a reconciliation of mixed findings.

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The possibility that governance devices exert differing effects in early vs. later stages of CEO tenure is important not only conceptually, but also practically. Owing to growing concern about managerial mischief, rules and standards of corporate governance have proliferated in recent decades, under the assumption that controls and incentives would foster R&D spending. But extant research has challenged this idea (Driver and Guedes, 2012; Devers et al., 2008); hence it is still an open question whether and how R&D investment can be enhanced by governance devices. And because the current economic downturn may give rise to new regulations and prescriptions (SEC, 2010, 2011; Barker, 2014), there is an urgent need to clarify the conditions under which these devices are more likely to be effective.

We contend that the two agency models of limited competence and managerial opportunism apply to early vs. later stages of CEO tenure respectively. We predict that stock options and board independence increase R&D investment later in CEO tenure but decrease it early in CEO tenure. An empirical test on 310 public companies in United States supports these hypothesized effects. In the next section, we examine how the two models relate to CEO tenure. A following section develops hypotheses on how stock options and board independence affect R&D spending in differing ways along the CEO tenure cycle. Methods and results sections describe an empirical test and a final section discusses implications for decisions on R&D investment and for policy makers.

2. Theory and background

2.1. R&D investment and corporate governance

Decisions on R&D investment are critical not only for firm success, but also for executives, owing to their peculiar characteristics. First, R&D investment is risky, as it may not result in any payoff (Lee and O'Neill, 2003; Driver and Guedes, 2012; Munari et al., 2010). Second, cash flows are asymmetrically distributed over time: payoffs materialize in the long run, whereas in the near term, R&D investment depresses performance (Driver and Guedes, 2012; Garcia-Manjon and Romero-Merino, 2012; Wang and Thornhill, 2010). Third, decisions on R&D investment are complex, because of ambiguous cause-effect relationships (McKenna, 1995; Munari et al., 2010), and are also difficult to communicate, because of the need for secrecy and protection of innovative ideas. Consequently, R&D investment has been a major focus for innovation and governance scholars, leading Munari et al. (2010, p. 1094) to affirm that decisions on R&D investment “are particularly indicative of the divergence between managers’ and shareholders’ interests” and “magnify the control problem generated by information asymmetries and potential moral hazard” (2010, p. 1094).

Most of this research on innovation effort draws from the agency model of managerial opportunism (Jensen and Meckling, 1976). According to this model, shareholders and CEOs exhibit different preferences for risky investment (Coffee, 1988): shareholders are risk neutral, because they can diversify their financial investment across multiple firms; CEOs are risk averse, because they cannot diversify their employment and compensation risk. As a result, CEOs likely take less risk than shareholders would prefer. Assuming that individuals maximize their own utility, CEOs may use their power to minimize R&D spending at the expense of shareholders (Ahuja et al., 2008); governance devices, such as stock options and independent boards of directors, are needed to prevent this underinvestment.

Stock options represent outcome-based compensation, which links the CEO's wealth to the firm's financial success; agency theorists contend that stock options boost R&D investment by aligning the CEO's interests with shareholders' financial returns. Board independence refers to the degree to which board

configuration favors an independent assessment of the CEO's proposals, and encompasses CEO duality and outsider ratio (Dalton et al., 2008). CEO duality/non-duality represent two alternative board configurations: CEO duality (i.e., the CEO is also the chair of the board) magnifies CEO power and compromises control, so that agency scholars prescribe separating the two roles (Krause et al., 2013). Outsider ratio refers to the proportion of independent directors on the board, that is, directors whose only tie with the CEO/organization is the directorship: a high level of outsider ratio strengthens control, because independent directors more freely side with shareholders when the CEO's interests are misaligned. Some scholars have expanded the analysis to other aspects of board configuration: for instance, in standard agency theory, monitoring committees facilitate board control, especially when boards are large (Reeb and Upadhyay, 2010; Upadhyay et al., 2014).

Extant research has sought to validate these ideas, but empirical evidence has been contradictory (Baysinger et al., 1991; Devers et al., 2008; Hill and Snell, 1988; Hoskisson et al., 2002; Kor, 2006; Zona et al., 2013). Several scholars have thus challenged the prediction of standard agency theory.

Among others, stewardship theorists challenged the “model of man” underlying agency theory (Hernandez, 2012). Building on motivational theories (Maslow, 1970; Argyris, 1973a,b), stewardship scholars suggest that CEOs behave as stewards, in the best interest of their principal and organization. Extending this line of argument, Hendry (2002) developed an alternative perspective on principal-agent relations, the agency model of limited competence. According to Hendry (2002), even setting aside the problem of managerial opportunism and assuming that CEOs behave like stewards (Davis et al., 1997),¹ agency problems can still arise owing to misinterpretation, misunderstandings, problems of communication, judgmental fallibility, limited foresight, and reduced task knowledge, phenomena that Hendry gathers together under the notion of “limited competence.”

In such conditions, different types of agency problems may arise. First, a principal's specification of objectives may bias the CEO's actions and behaviors to the detriment of the principal itself: a CEO will seek to attain objectives specified in advance to appease the principal's expectations, even when – under the evolving circumstances and the “limited competence” inherent in the management situation – these objectives become obsolete and/or misaligned with the principal's real interests. Second, direct guidance by the principal, via close interpersonal interactions with the agent, restricts a CEO's free exercise of discretion and dampens the opportunity to benefit from his talent. In both cases, the principal bears agency costs, i.e., deviation of agent's outcomes from the principal's real interests (2002, p. 102). Under limited competence, governance devices may exert effects very different from those envisaged by standard agency theory. For example, specialized committees, with their competence, may help alleviate the problem of ‘limited competence’ (Lejarraga and Martinez-Ros, 2014), improving board functioning and supporting CEOs to increase R&D investment.

The agency model of limited competence has received scant attention, despite its ability to broaden the scholarly view of agency problems. We contend that the determinants of R&D spending can be best understood by applying both agency models,

¹ In the words of Hendry (2002): “I assume that both principals and agents are honest and that, having entered into obligations to achieve, as best they can, their principals' objectives, agent seek to dutifully do that. I assume, in other words, that agents behave like stewards in the Donaldson's original sense of the term” (Hendry, 2002, p. 101). The model of limited competence offers an extended perspective to frame the role of outcome-based compensation and to specify coordination problems under limited competence (which arise early in CEO tenure).

because each captures some aspects of principal-agent relations in differing situations. The two agency models of managerial opportunism and limited competence are complementary: “managers, like anyone else, exhibit a mixture of motivations and behaviors. In management situations in which the problems arising from incompetence are of greater concern than those arising from opportunism, [the agency model of limited competence] provides a more plausible alternative to the standard theory” (Hendry, 2002, p. 109). In the next section we argue that the two models prevail at differing stages of CEO tenure; then, we develop hypotheses on how R&D investment may be increased or decreased by governance devices, in early vs. later stages of CEO tenure.

2.2. Agency models along the CEO tenure cycle

CEOs’ psychological biases influence the firm’s innovation effort (Ahuja et al., 2008, pp. 61–62; Lee and O’Neill, 2003). Specifically, a CEO’s interests, motivations, and power evolve along his/her tenure cycle (Hambrick and Fukutomi, 1991), so that the proclivity to invest in R&D may be more or less pronounced in early vs. later stages of CEO tenure (Ahuja et al., 2008). As we argue below, the agency models of limited competence and managerial opportunism most likely apply at differing stages of CEO tenure.

2.2.1. Early stage of CEO tenure

In the early stage of a CEO’s tenure, the agency problem of limited competence prevails and significantly shapes principal-agent relations. Both the board and the CEO suffer from limited competence, and experience problems of communication, misinterpretation, misunderstanding, missteps, judgmental fallibility, limited foresight, and reduced task-knowledge.

As regards the CEO: “a new CEO enters the job at a disadvantage in terms of knowledge of the task – the facts, trends, contacts and procedures that pertain to the successful conduct of the CEO’s role in the focal firm” (Hambrick and Fukutomi, 1991, p. 725); new CEOs have not yet developed – together with subordinates and staff – the format, timing and even the content of information exchanges (Hambrick and Fukutomi, 1991), leading to problems of communications and misunderstanding. In an attempt to succeed, new CEOs engage in learning and experimentation (Miller and Shamsie, 2001), and incur all the by-products of learning, such as misunderstandings, missteps, and misinterpretations. Indeed, “novice CEOs tend to make many mistakes” (Miller and Shamsie, 2001, p. 737), since they are too inexperienced about their companies and environments to choose the right strategy (2001, p. 728). These problems of limited competence become salient when it comes to R&D investment, given the inherent complexity and high-risk profile of such decisions. Lack of experience, as CEO in the focal firm, increases the magnitude of actual (or perceived) losses associated with risky R&D investment (Simsek, 2007, p. 654); it may be difficult for new CEOs to comprehensively assess and justify R&D investment *vis-à-vis* the board (2007, p. 654).

As regards the board: in early years of a CEO’s tenure, the board’s major task consists of assessing the suitability of the new CEO, but directors perform such a task under limited competence. “When hiring CEOs, boards make their best, but inevitably imprecise, estimates of the CEOs’ talent and motivation” (Wowak et al., 2012, p. 721), so that “a CEO succession process does not end when a new CEO is appointed; it can be a continuous evaluation process that lasts for a few years after the succession” (Zhang, 2008, p. 861). CEO ability becomes clear over time (e.g., Fama, 1980): “the principal (the board) can learn about the agent’s (new CEO) ability over time and the updating of the ability estimate becomes more informative with each successive period” (Zhang, 2008, p. 861). Thus, boards and shareholders cannot predict or assess with any

confidence a CEO’s suitability for the corporation, nor responses to power struggles that often accompany succession.

Whereas limited competence affects both the CEO and the board, early in CEO tenure the agency problem of managerial opportunism is attenuated (Shen, 2003). New CEOs are mostly concerned to achieve legitimacy and secure their job, and to this aim seek to meet the principal’s expectations; they are under close scrutiny of stakeholders, boards and subordinates (Shen and Cannella, 2002), and experience high threat of dismissal (Khurana, 2001; Ocasio, 1994). They lack inside political footholds and depend for information on managers (Shen and Cannella, 2002). New CEOs are thus unlikely to engage in managerial opportunism, because such behavior would threaten their position.

2.2.2. Later stages of CEO tenure

In later stages of their tenure, the agency problem of managerial opportunism becomes salient. First, long-tenured CEOs have interests that most likely depart from those of shareholders: they experience low task interest and a reduced inclination to take risk (Hambrick and Fukutomi, 1991; Miller and Shamsie, 2001), and may thus “have little interest in pursuing strategies of innovation through higher R&D spending, preferring instead to emphasize stability and efficiency” (Barker and Mueller, 2002, p. 784), expanding the risk-differential between the CEO and the shareholders. Second, long-tenured CEOs accumulate power (Finkelstein, 1992), and are able to pursue their interests opportunistically, and reduce R&D spending.

Whereas the agency problem of managerial opportunism is magnified, the agency problem of limited competence is mitigated in late stages: CEOs accumulate task knowledge and an intimate understanding of the firm (Miller and Shamsie, 2001); boards and shareholders accumulate awareness of CEOs’ skills and management styles, so that CEOs become predictable (Hambrick and Fukutomi, 1991); a CEO’s corporate strategy has been identified and legitimated, so that shareholders consolidate their expectations. Moreover, the CEO’s job security is no longer threatened: the CEO’s leadership is taken for granted and her/his authority is no longer questioned, as long as his/her performance is acceptable (Hambrick and Fukutomi, 1991).

In sum, whereas early in CEO tenure the agency problem of limited competence prevails, in later stages of tenure the problem of managerial opportunism becomes salient. In the next section, we develop hypotheses on how stock options and board independence shape R&D investment in these differing stages.

3. Hypothesis development

3.1. R&D investment and the effect of CEO stock options in early vs. later stages of CEO tenure

Agency scholars have developed the “incentive-based view” of stock options to suggest that stock options increase R&D spending: by linking CEO pay to financial performance, they make a CEO “eligible to participate in future gains of a company’s share price” (Sanders and Hambrick, 2007, p. 1060). Hence, stock options enhance a CEO’s motivation to allocate resources to risky projects that may substantially increase financial performance in the long run, such as R&D investment.

An emerging, opposite view, the “behavioral agency view,” suggests that stock options reduce R&D spending (Devers et al., 2008). Building on prospect theory (Kahneman and Tversky, 1979), this view argues that CEOs are concerned to protect their existing wealth against potential losses, rather than increasing it in the future, and will avoid risky actions (R&D investment) that may

threaten that wealth.² Specifically, CEOs ‘instantly endow’ to their wealth their current and future (expected) stock options, including both exercisable (Devers et al., 2008) and un-exercisable (Larrazza-Kintana et al., 2007) options: they perceive option value as real wealth they already own, and seek to protect that wealth by behaving more conservatively. Thus, the larger the stock options, the greater a CEO’s perceived wealth-at-risk-of-loss, and the less his/her willingness to invest in risky R&D (Wiseman and Gomez-Mejia, 1998).

We contend that the two agency views prevail at differing stages of CEO tenure: in early stages of CEO tenure, stock options reduce R&D spending; in later stages, they increase it.

Stock options and R&D investment in the early stage of CEO tenure. In this stage, the “incentive-based view” hardly applies: the opportunities for wealth increase materialize only in the long run,³ well beyond the time-horizon of new CEOs, who experience highest rate of dismissal (Hambrick and Fukutomi, 1991) and focus on securing their job in the near term (Shen, 2003). As we shall argue below, large stock options will reduce R&D investment, because they (a) aggravate this perceived risk and (b) bias a CEO’s problem framing.

First, large option holdings aggravate a CEO risk bearing, reducing his/her proclivity to propose risky actions to the board: in accordance with the “behavioral agency view”, under large option holdings, not only do new CEOs fear losing their job; they also fear losing option wealth associated with it.⁴ Thus, the larger their stock options, the larger the perceived wealth-at-risk-of-loss, and the less their R&D investment.⁵ Indeed, scholars in the “behavioral agency view” have argued that when CEO tenure is at risk and CEOs have large option holdings, CEOs “maximize the easier to attain at the expense of more uncertain long-term returns, such as those in R&D investment” (Makri et al., 2006, p. 1059): reducing R&D investment improves firm performance immediately, enabling CEOs to demonstrate their superior ability (Deutsch, 2007); Laux (2012) outlines how, when a CEO is subject to potential replacement, stock options may backfire and encourage short-term investments: a CEO is concerned about forfeiting stock options due to dismissal, and boosts short-term performance, in order to improve a board’s perception of his/her ability (2012, p. 522).

Second, stock options will alter a CEO’s *framing and interpretation* of the situation, biasing decision making. Under limited competence, CEOs have unclear and ambiguous understanding of the principal’s preferences; they seize any cue to interpret board/shareholders’ expectations (Hendry, 2002). As outlined by Hendry (2002), in such conditions stock options “convey information on the principal wishes” (2002, p. 103), and “specify a principal’s objectives in a particularly clear and precise way” (Hendry, 2002, p. 103), placing greater emphasis on financial performance; in Hendry’s words, the more CEO pay depends on meeting financial objectives inherent in stock options, “the more

rigidly the specification may be interpreted as defining the principal’s objectives” (p. 103), leading CEOs to place greater emphasis on financial prospects as a criterion to select initiatives to the board. Yet, an emphasis on financial prospects and controls reduces R&D expenditures (Baysinger and Hoskisson, 1989); a new CEO likely restricts R&D budgets to projects that appear more justifiable from a financial perspective, and will discard exploratory projects that – at very early stages of development – are still difficult to defend in terms of future financial prospects. By contrast, if compensation is in no way dependent on financial objectives, this fact will be interpreted as giving the agent license to modify, adapt, or reinterpret them to meet new circumstances (see Hendry, 2002). Such a freedom is essential for R&D investment, because innovative projects seldom manifest their full financial potential in advance, and require long periods of experimentation, learning, and support in the face of initial failure. A CEO can more willingly devote resources to R&D initiatives if he/she believes that reconsideration/discarding of previously approved initiatives will not lead the shareholders/board to blame her/him for having wasted those resources. In sum, assuming that CEOs are boundedly rational (Hendry, 2002), under ambiguity of ‘limited competence’, the presence of stock options makes a major difference in a CEO’s problem framing, emphasizing financial prospects and restricting R&D budgets.

Stock options in later stages of CEO tenure. In later stages of tenure, large stock options boost R&D spending: the “incentive view of stock options” predominates. The CEO’s task becomes less novel and therefore less intrinsically interesting, and CEO motivations decouple from those of the organization/shareholders (Hambrick and Fukutomi, 1991); moreover, accumulated power gives longer-tenured CEOs more room to act in their own self-interest at the expense of shareholders (Barker and Mueller, 2002). In the absence of stock options, CEOs are likely to devote resources to consumption of perquisites, to pet projects, and to utility-enhancing initiatives, reducing resources allocated to R&D. At the same time, the “behavioral agency view” is less applicable: stock options are less likely to trigger risk bearing, since long-tenured CEOs may use their power to reload or reprice stock options, and avoid threat to personal wealth, inherent in undertaking risky projects such as R&D investment (Devers et al., 2008).

This argument is also consistent with the agency model of limited competence: as CEO tenure progresses, the limited competence attenuates. Shareholders and boards learn the CEO’s corporate strategy: since “CEOs become predictable” (Hambrick and Fukutomi, 1991), boards can precisely formulate expected outcomes and communicate them to CEOs. As Hendry explains, when the limited competence attenuates, and objectives stated in advance more accurately reflect the principal’s *real* objectives, the specification of objectives is beneficial, as it helps focus agents to attain the specified targets (Hendry, 2002, p. 104). Hence,

Hypothesis 1. Higher levels of CEO stock options are associated with lower levels of R&D investment in firms with CEOs at early stages of their tenure, and with higher levels of R&D investment in firms with CEOs at later stages of their tenure.

3.2. Board independence and R&D investment

We contend that board independence reduces R&D investment in the early stage of CEO tenure, but increases it in later stages. “Early-stage CEO evaluation differs significantly from CEO evaluation later in their tenure” (Graffin et al., 2013, p. 399), due to the “extreme evaluative uncertainty associated with CEOs’ assessment early in their tenure” (2013, p. 397). Evaluative uncertainty refers to the absence of clear and unambiguous indicators or benchmarks of performance (Graffin and Ward, 2010): early in a CEO

² R&D spending may be totally unproductive, resulting in share price depreciation. Financial returns materialize in the long-term, but in the near term, R&D spending entails a sure increase in costs and a decrease in financial performance.

³ When payoffs from R&D investment come into existence, and eventually translate into share price appreciation.

⁴ For example: to lose unvested stock options; or the opportunity for further option awards, under the rich pay plans at the focal firm, which wealth CEOs ‘instantly endow’ to their current wealth. In the behavioral agency model, ‘instant endowment’ refers to the human beings’ tendency to frame a “possibility” of future wealth, as wealth already own, altering behaviors to reduce risky investment (see Wiseman and Gomez-Mejia, 1998; Devers et al., 2008).

⁵ Such a bias will occur also under the assumption of a CEO/steward, because – Hendry states – under limited competence the CEO may be unable to assess with any confidence which action (e.g., investing vs. not investing in an innovative initiative) best serves the principal, and financial incentives bias the choice in favor of the alternative that also increase the CEO’s interest, with no intention of harming the principal.

tenure, firm performance may be at best a noisy indicator of the CEO's talent (Grafflin et al., 2013, p. 384); "objective performance metrics are not readily available" and "directors will be forced to make *subjective* judgments of CEO quality" (2013, p. 388). This point is in accordance with Hendry's idea that, under ambiguity, the principal assesses the agent by relying on subjective criteria, such as behaviors and personal interactions: "principals are likely to communicate through informal dialog, using examples and generalizations" (2002, p. 101); "the ability of agents to meet their principals' objectives is a function... of the extent to which they are able to *empathize* with the principal" (2002, p. 102).⁶

But subjective valuation by the board exacerbates a CEO's risk bearing,⁷ and hence reduces risk taking and R&D investment. According to Wiseman and Gomez-Mejia (1998), under subjective valuation, a CEO has little or no cue in advance about how he/she will be evaluated *ex post*, and evaluation will depend strongly on personal interactions with directors. Valuation based on behaviors is subject to biases because behavior is, in itself, equivocal and subject to unique interpretation that is experience dependent (Weick, 1970). Thus, "if an agent has reason to suspect that taking risks may lead to negative evaluations because the appraisal criteria and performance targets are ill defined until after the actions are taken – and are thus subject to a wide range of hindsight interpretation – then the agent should frame the situation as riskier and should reduce risk taking" (Wiseman and Gomez-Mejia, 1998, p. 145).

We contend that, in such conditions, board independence magnifies CEO risk bearing and reduces risk taking – and hence R&D investment – to a greater extent, whereas board dependence mitigates risk bearing and fosters R&D investment; board independence may become beneficial only in later stages of tenure. We examine this idea in reference to two aspects of board independence: the CEO duality/non duality structure, and outsider ratio (Dalton et al., 2008).

3.2.1. R&D investment and CEO duality in early vs. later stages of CEO tenure

Standard agency theorists prescribe nonduality (separating the roles of CEO and board chair) as a defense against opportunism (Baliga et al., 1996). We argue that this arrangement works in later stages of CEO tenure, but in the early stage, nonduality may reduce R&D expenditures. Under conditions of limited competence (i.e., early in CEO tenure), nonduality nurtures rivalry with a powerful chair, ambiguous relations with subordinates, and an inability to face pressures from stakeholders, accentuating risk bearing (perceived risk) and reducing a CEO's inclination to invest in risky R&D projects.⁸

⁶ Zhang (2008, p. 861) clarifies that, in the early years of CEO tenure, "the board can update its ability estimate of the new CEO, as it observes and evaluates how the new CEO applies his or her existing knowledge and skills and develops new task knowledge and skills" (Zhang, 2008, p. 861). Gabarro also outlines how a decision to retain/dismiss a new CEO depends on ongoing interpersonal interactions between a CEO and the board: "success is influenced by a number of personal and interpersonal factors, such as management style, interpersonal skills, leadership ability and the potential for conflicts in management style" (Gabarro, 1987, p. 68). Even the mandate, usually more implicit than explicit, contains a general message concerning the magnitude and direction of change, so that behaviors and interactions ultimately determine the board's decision to retain/dismiss a new CEO (Hambrick and Fukutomi, 1991).

⁷ Risk bearing refers to the "perceived risk to an agent's wealth, that can result from employment risk or other threat to wealth" (Wiseman and Gomez-Mejia, 1998, p. 136), and refers to the degree to which the CEO perceives the situation as risky or threatening his/her wealth/employment. The higher the risk bearing, the lower the risk taking.

⁸ Stewardship theory would argue that non-duality impairs CEO motivation. The agency model of limited competence points to problems of coordination and of interactions between the principal and agent.

First, early in CEO tenure, nonduality leads to potential conflict between the new CEO and a powerful incumbent chair. Gabarro (1987, p. 60) outlines that newly appointed executives frequently face interpersonal problems that go beyond task-related conflict and stem "from less rational factors": nonduality loosens unity of command at the top, creating multiple authority relationships that promote role ambiguity, uncertainty, misunderstanding, and potential personal conflicts between the CEO and the chair (Finkelstein and D'Aveni, 1994). These conflicts reduce a new CEO's ability to succeed, and by increasing perceived employment risk (Wiseman and Gomez-Mejia, 1998), they reduce R&D spending. In other words, it becomes more risky for a new CEO to propose uncertain initiatives and expand R&D budgets, when a separate person chairs the board: early in a CEO tenure, the board assesses a CEO quality; a new CEO will struggle to demonstrate competence and ability, but he/she needs to do so under ambiguity and uncertainty: the presence of a separate Chair boosts evaluative pressure and reduces a CEO proclivity to propose projects more difficult to justify, restricting R&D budgets. Second, nonduality also shapes the relation of a new CEO with subordinate executives: it creates ambiguity among subordinates about the existing lines of authority and affects how they respond to change and innovation initiatives undertaken by the newly appointed CEO, thus weakening his/her ability to take charge (Finkelstein and D'Aveni, 1994) and to learn and experiment (Gabarro, 1987, p. 60): top executives feel encouraged to challenge a new CEO's proposals, when a separate Chair is present and may listen to their criticism. Third, nonduality also weakens a new CEO's ability to succeed in the face of stakeholders' pressures, because, by engendering ambiguity and diffusion of power at the top (Finkelstein and D'Aveni, 1994, p. 1083), it renders decisive action more difficult (Miller and Friesen, 1977, p. 268), just when such actions are particularly warranted to meet stakeholders' expectations for a quick corporate turnaround. In contrast, CEO duality gives newly appointed CEOs greater power and control over the board, subordinates, and stakeholders, reducing ambiguity and uncertainty, mitigating risk bearing and leading to greater R&D investment (Finkelstein and D'Aveni, 1994).

All these arguments hold in early stages of CEO tenure. However, in later stages of tenure, CEO power and managerial opportunism prevail: the dominant problem becomes one of excessive CEO power and the risk of shareholders' expropriation. CEO duality may exacerbate this problem, leading to reduced R&D investment. Under CEO duality, "the CEO will acquire a wider power base and locus of control, thereby weakening decision control by the board" (Boyd, 1995, p. 303). Jensen (1993) argued that the CEO duality compromises the board's ability to monitor the CEO: "without the direction of an independent leader, it is much more difficult for the board to perform its critical function" (Jensen, 1993, p. 866). Under CEO duality, directors may be unable or unwilling to dispassionately evaluate the CEO's policies and practices (Dalton et al., 2008). Indeed, CEO duality has been equated with the "CEO grading his own homework" (Brickley et al., 1997, p. 190). With this stronger power-base, a long-tenured CEO may more easily pursue his own agenda, exactly when it may differ substantially from shareholders' goals. By contrast, separating the CEO and chair positions would strengthen monitoring and reduce managerial opportunism: "the choice to introduce an independent voice with structural power over a CEO (i.e., a separate Chair) into board discussions can lead to more critical reviews of the CEO's strategies, and ideally to more shareholder-focused strategic action" (Krause and Semadeni, 2013, p. 805), such as greater R&D expenditures.

Hypothesis 2. CEO duality is associated with higher levels of R&D investment in firms with CEOs at early stages of their tenure, and with lower levels of R&D investment in firms with CEOs at later stages of their tenure.

3.2.2. R&D investment and outsider ratio in early vs. later stages of CEO tenure

We argue that outsider ratio, like duality, will exert differing effects on R&D spending in early vs. later stages of CEO tenure. Early in CEO tenure, a high outsider ratio will reduce risky R&D spending, because it enhances behavioral control and the usage of decision-making subjective heuristics by directors. The CEO and outside directors have limited opportunities to work closely, increasing the ambiguity and uncertainty that typically characterize behavioral control; outsiders have little day-to-day interaction with the newly appointed CEO, and will rely more extensively on decision-making heuristics to assess a new CEO, increasing the CEO's risk bearing.

Succeeding at the outset of tenure “requires an ongoing investment of time on the part of the new manager in meeting with key subordinates and superiors to work out priorities and settle differences” (Gabarro, 1987, p. 81). Specifically, Gabarro (1987) states that a key factor that distinguishes successful from unsuccessful managerial turnover is the new manager's effectiveness in working through a set of shared expectations about goals, priorities, and roles with key subordinates and superiors: but “developing a larger sense of shared purposes and cohesion takes considerably longer in the successful cases” (p. 81); “much of the actual work of developing a set of *shared expectations* about how the organization should function occurs with more modest issues. In many cases, the reviews that are a normal part of managing a business provide natural arenas in which this work is accomplished” (p. 83).

Board meetings occur only episodically (Forbes and Milliken, 1999) and represent major milestones in adjustments of the CEO's and directors' preferences and expectations. Since outside directors do not work in the firm, a high outsider ratio diminishes the possibility for a new CEO to work out differences and shared expectations with directors on the “natural,” gradual basis described above. Hence, the greater the proportion of outside directors, the more intensely the new CEO perceives ambiguity and uncertainty about her/his own evaluation, as he/she waits for the next board meeting to frame prospects and expectations that have not been adjusted since the previous meeting. Thus, the greater a CEO risk bearing about board evaluative criteria, the less his/her proclivity to invest in risky R&D initiatives.

Two more arguments support our prediction. First, as Halebian and Rajagopalan note (2006, p. 1016), outside directors are likelier to “make internal attributions for poor performance and blame CEOs for performance problems beyond their control in order to placate powerful stakeholders.” This is particularly detrimental early in CEO tenure, when stakeholders' pressures for quick turnaround and visible results are most intense. Second, when *inside* directors predominate on the board, a CEO can more easily limit negative performance appraisal by gaining preliminary acceptance (“buy-in”) from inside board members for strategic choices: this tactic “ultimately reduces the ambiguity associated with the use of behavioral evaluation criteria, since boards would have expressed commitment to specific behaviors in advance” (Wiseman and Gomez-Mejia, 1998, p. 148). The greater the portion of outside directors on the board, the less a new CEO can engage in “buy-in” tactics, so he/she bears greater risk and takes less.

Yet in later stages of CEO tenure a greater outsider ratio may be beneficial: a greater proportion of independent directors on the board may be particularly suited to curb managerial opportunism and enhance R&D spending when the problems of excess CEO power and managerial opportunism are most salient. Non-independent executive directors report directly to the CEO, and depend on him/her for pay raises, promotions, career opportunities, task assignments, and other benefits; they are likely to avoid challenging the CEO for fear of retaliation (Dalton et al., 2008), especially

when the CEO has accumulated significant power and legitimacy inside and outside the organization. Non-independent directors who have business links with the firm are similarly exposed to retaliation (e.g., the CEO may not renew the consulting contract). By contrast, independent directors are more likely to challenge the CEO and preserve shareholders' interests. Hence, later in CEO tenure, a larger portion of independent outside directors on the board curbs managerial opportunism and enhances R&D investments.

Hypothesis 3. Higher levels of outsider ratio are associated with lower levels of R&D investment in firms with CEOs at early stages of their tenure, and with higher levels of R&D investment in firms with CEOs at later stages of their tenure.

4. Methods

4.1. Sample and data

The empirical analyses were carried out on a panel dataset of 310 public companies in the United States, from 2001 through 2007, for 2170 firm-year observations. Data were collected from multiple sources: Compustat, Osiris, Corporate Library, and Execucomp. The sample excluded firms that reported zero values for R&D expenditures. For this time window, firms included a significant number of CEOs in their early stages of CEO tenure: in 149 firms, CEOs were constantly at the helm in the first three years; in 161 firms, CEOs had been in office for at least three years in later stages of tenure. The evidence is in accord with research showing how CEOs are frequently dismissed early in their tenure (Miller and Shamsie, 2001).

We test our hypotheses by following the same approach by Hou et al. (2014): in examining the relation between CEO tenure and performance, these scholars adopt two distinct and complementary specifications: a 3-year cut-off specification, to distinguish between early and late stages of CEO tenure, and a continuous, quadratic specification of CEO tenure. As outlined by Hou et al. (2014), according to Hambrick and Fukutomi (1991), in the first three years of tenure, new CEOs are likely in their “response to mandate” and “experimentation” stages (Hou et al., 2014, p. 11). The three-year cut-off specification is also based on extensive research on CEO tenure and dismissal of newly appointed CEOs: scholars recognize the dismissal of new CEOs as a core research topic for management, given the disruptive consequences for corporations and the inherent waste of executive talent (Khurana, 2001; Shen, 2003). Research in this field generally adopts the 3-year cut-off specification to distinguish CEOs in their early stages of tenure.⁹ Indeed, “a newly appointed CEO begins with a period of extreme vulnerability; this early vulnerability is reflected in the disproportionate number of CEOs whose tenure last three years or less” (Fredrickson et al., 1988, p. 258): this phenomenon is confirmed by the fact that almost half of the total number of CEO dismissals occurs within the first three years in office (Zhang, 2008). Hence, Finkelstein and Hambrick (1996) explicitly suggest the first 3 years of tenure as the appropriate time-window to study the survival of new CEOs. The extreme vulnerability and the high-rate of dismissal of new CEOs is related to our arguments: the problem of limited competence – which we argue is aggravated by governance structures – is salient for new CEOs, and influences decision making

⁹ For example, the 3-year cut-off specification has been used: by Zhang (2008) in studying dismissal during the early years of CEO tenure; by Gabarro (1987) in studying the dynamics of newly appointed managers; by Zhang and Rajagopalan (2004) in their study on how CEO succession influences organizational outcomes; by Barker and Mueller (2002) in their study on CEO tenure and R&D investment; by Zhang and Rajagopalan (2010) in their study on CEO origin and strategic change.

in R&D, because of the inherent implications of misunderstanding, mistakes, misinterpretations etc. for their survival on the job and for their wealth. Hence, we expect our hypothesized effects to be especially significant in the first three years of tenure, i.e., when new CEOs experience the greatest vulnerability on the job. Nevertheless, the phenomena of limited competence and managerial opportunism are likely to manifest in a continuous process, because both learning and power accumulate over time. Therefore, we also adopt a quadratic, continuous specification (Hambrick and Fukutomi, 1991) to account for these ongoing processes.

4.2. Dependent variable

We operationalize R&D investment as *R&D intensity*, i.e., measured as the ratio of R&D investment divided by total assets (Kor, 2006). The operationalization of R&D intensity has been employed in investment studies (Mairesse et al., 1999; Bond and Van Reenen, 2007); used with dynamic models for panel data, it has the advantages of making it more likely that the dependent variable is stationary (Driver and Guedes, 2012) and of modeling explicitly cyclical adjustment (Harhoff, 1998).

4.3. Independent variables

We measured *CEO stock options* as the logarithmic transformation of the dollar value of option awards to CEOs, in accordance with previous studies (e.g., Sanders and Hambrick, 2007). We measured *CEO duality* as a dummy variable that equals 1 for firms where the roles of the CEO and the chair are combined. We measured *Outsider ratio* as the ratio of independent outside directors to all directors. We measured *Early CEO tenure* as a dummy variable (1 = first three years in office) for the 3-year cut-off operationalization, and a continuous measure of *CEO tenure* for the continuous quadratic operationalization.

4.4. Control variables

We control for past investments, by including a *one-year lagged dependent variable*. We included several control variables. The variable *Technological opportunities*, measured as the average ratio of R&D industry expenditures to total sales (Tihanyi et al., 2003), accounts for industries whose competition is heavily based on developing new products and technologies, and hence on greater R&D spending (Geroski, 1990; Zahra & Covin, 1995). *Industry Appropriability*, measured as “the total number of industry patents per industry over total number of companies that patent in that industry” (Artés, 2009, p. 124; Beneito, 2003), controls for the firms’ ability to capture the rents from innovation (Teece, 1986), fostering the motivation toward greater R&D investment (Encaoua et al., 2006; Gallini, 2002). *Industry concentration*, measured as the Herfindal index of competitors’ market shares in a given industry (2 digit SIC codes) (Ahuja et al., 2008), accounts for market power and concentration, which are believed to boost innovation effort (Cohen and Levin, 1989), though some studies report the opposite (Ahuja et al., 2008).

At the firm level, we control for *Firm size*, measured as the logarithmic transformation of total sales: larger firm size facilitates innovation success and fosters the motivation to invest in R&D (Damanpour, 2010; Schumpeter, 1942). *Firm age*, measured as number of years since the firm’s foundation, is included since younger firms experience liability of newness in exploring new R&D investment (Stinchcombe, 1965). *Ownership concentration* (Jensen and Meckling, 1976), measured as the percentage of ownership held by blockholders having more than 5% of the firm equity, reduces the agency problem of underinvestment and raises R&D investment (Hoskisson et al., 2002). *Firm performance* is measured

as return on equity (Li and Tang, 2010): it has been found that under declining performance, managers invest more in R&D as a means to revert the trend (Greve, 2003); other studies propose an opposite view and the effect is not univocal (Mone et al., 1998; Nohria & Gulati, 1996).

We control for components of CEO pay, such as the dollar value of *CEO salary* and *CEO bonus*, since previous studies have shown that such components of pay enhance R&D investment (Balkin et al., 2000; Makri et al., 2006). *CEO scientific background* (with a dummy variable that equals “1” for CEOs having a background in R&D) is included, since CEOs with such a background exhibit a greater propensity to invest in R&D (Johnson et al., 2013; Barker and Mueller, 2002). We control for *Board size*, measured as the number of directors on the board: board size may impair R&D investment, because larger groups exhibit a reduced propensity to take risk, due to the need of intra-group compromise among group members (Sah and Stiglitz, 1986, 1991; Zona et al., 2013; O’Reilly et al., 1989); however, it may also result in greater resources and support available from the external environment, fostering R&D investment (Hillman et al., 2009). We also control for *Board committees*, by including a dummy variable which equals 1 for the presence of more than 3 committees: this measure has been used (Upadhyay et al., 2014) to control for the possibility that improved division of labor at the board level would enhance effectiveness, reduce the agency problem and foster R&D investment. While we assess *CEO stock options*, *CEO duality* and *Outsider ratio*, in early vs. later stages of CEO tenure, as our core predictors, we also include them as controls in the basic model (Kor, 2006; Driver and Guedes, 2012); specifically, outsider ratio may enhance R&D spending, by providing competence/resources from the external environment (Hillman et al., 2009). Finally, we included *industry dummies* (2 digit US SIC-codes), as firms’ propensities toward R&D investment vary across industries (Ito and Pucik, 1993), and *year dummies* to account for time effects.

4.5. Econometric method

To estimate our model we used the system GMM estimator (Roodman, 2009), which allows us to deal with multiple problems, such as endogeneity, simultaneity, the potential correlation of time-invariant firm characteristics (fixed effects) with explanatory variables; the fixed effects contained in the error term, which consist of unobserved firm-specific effects and observation-specific errors; potential autocorrelation due to the lagged dependent variable; and the presence of unobserved shocks in the firm’s fixed effect that show in the error term, which may bias regression results in small-T datasets. The GMM estimator deals with these issues: (a) it accounts for endogeneity by drawing instrumental variables from the dataset itself; (b) it accounts for time-invariant firm characteristics by using first differences to transform regressors and removing any fixed firm-specific effect; (c) it deals with a first-differenced lagged dependent variable by instrumenting its values; and (d) it adjusts for any unobserved shock in the firm’s fixed effect that might distort the results in small-T datasets, (e) and for simultaneity by constructing instruments from the past observations of variables. This methodology also accounts for variables not included in the model: these variables “are subsumed in the fixed effects term that includes all non-time varying determinants” (Driver and Guedes, 2012, p. 1570). The GMM estimator yields Hansen tests (Hansen, 1992), with a null hypothesis that specified variables are appropriate instruments: a non-significant test suggests instruments are valid. The Hansen test can also be viewed as a test of structural specification: omitting important explanatory variables would move components of variation into the error term making them correlated with the instruments: a non-significance test indicates that models do not

suffer from omitted variable problems. Independent variables are standardized and lagged one year. In interpreting results, it is worth outlining that controlling for a lagged dependent variable generally catches a large proportion of variance to be explained by other factors: thus, coefficients of control and independent variables are often small-sized and non-significant (Achen, 2001): a significance at 5% should thus be regarded as strong support.

5. Results

5.1. Main results

Table 1 displays descriptive statistics and correlations. The mean value of R&D intensity is 0.05: on average, R&D expenditures represent 5% of total assets. The standard deviation is 0.05, indicating that 95% of sample firms ranges from zero and 0.15 (i.e., mean ± 2 standard deviations). The ratio becomes especially higher for relatively small and young firms. As regard independent variables: the average outsider ratio is .79, with a s.d. of .10: hence, for the vast majority of sample-firms, outsider ratio is greater than .50 (i.e., the range of mean ± 2 s.d., that is, a range from .59 through .99, includes 95% of sample firms). The mean value of CEO stock options amounts to 2.840 US\$/000, with a standard deviation of 3.585 US\$/000; the mean value of CEO duality is 0.64 (s.d. = .40), indicating that 64% of firm-year observations exhibit a combined CEO-Chair governance structure. Table 1 also provides information on other governance variables: the average board size is 11 members: given the small standard deviation (s.d. = 0.94), most firms are concentrated around the mean, i.e., between 9 and 13 members; the mean of the dummy variable Board committees is 0.76: in 76% of sample-observations there more than 3 committees. Table 1 also gives information on pairwise correlations. The negative correlations of R&D intensity with firm size ($r = -.22$) and with Firm age ($r = -.28$) indicate that R&D intensity is greater in small-sized and younger firms. R&D intensity is also negatively correlated with Industry concentration ($r = -.26$). R&D intensity is positively correlated with CEO scientific background ($r = .22$), suggesting that CEOs with such a background are associated with more R&D-intensive corporations; R&D intensity is negatively correlated with CEO salary ($r = -.22$), suggesting that fixed compensation is higher in less R&D-intensive firms. There are relatively weak correlations of R&D intensity and independent variables, such as CEO stock option, CEO duality and outsider ratio; this result is consistent with our work: according to our hypotheses, independent variables exert opposite effects (positive vs. negative) along the CEO tenure; since the sample is approximately split in half, between early and late CEO tenure, the positive and negative effects tend to off-set one another. Finally, Table 1 provides information on correlations of other governance variables: outsider ratio is more strongly and positively correlated with CEO salary ($r = .23$), with a relatively large number of Board committees ($r = .31$), and with CEO duality ($r = .21$); the variable CEO stock options is more strongly related with CEO bonus (.13), suggesting that firms adopting incentive schemes, pay more in both forms of compensation. Moreover, the positive correlation of CEO stock options and firm size ($r = .19$) suggests that CEOs gain larger amounts of stock option in larger firms; the negative correlation of CEO stock options and Ownership concentration, suggests that CEO are granted larger amounts of stock options in firms whose ownership is more dispersed. Finally, CEO duality is more strongly correlated with CEO salary ($r = .25$) and Board committee ($r = .14$), suggesting that the combined structures tend to be associated with greater levels of fixed compensation and boards with a larger number of committees.

Table 2 reports regression results for the 3-year cut-off operationalization, whereas Table 3 reports the results for the quadratic, continuous specification. Hansen tests are all non-significant in

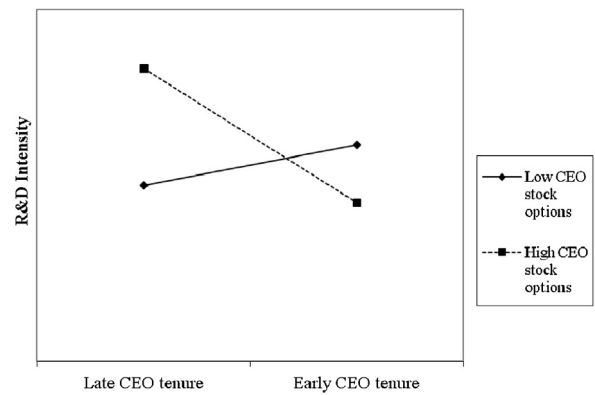


Fig. 1. Interaction effect: stages of CEO tenure X CEO stock option.

both tables: as outlined in Section 4.5, these tests suggest the validity of instruments and the absence of omitted variable problems. Model 2 of Table 2 reports that the interaction of CEO stock options and Early CEO tenure is negative and significant ($b = -.014$; $p < .05$), providing support for Hypothesis 1. The coefficient indicates that the CEO stock options reduce R&D intensity in early stages of CEO tenure, relative to late stages of CEO tenure. Since the variable is standardized, the coefficient represents the change in the dependent variable for a standard deviation change of the independent variable: hence, considering the interaction effects, in late stages of a CEO tenure, on average and holding other factors constant, CEOs with greater stock options for 3.585 mln US\$, will invest more in R&D for approx 1% in R&D intensity; by contrast, in early stage of CEO tenure, such a change in stock options entails a reduction of R&D intensity by 0.4%. Fig. 1 depicts the interaction effect, according to instructions by Aiken and West (1991): when the CEO is in the early stage of his/her tenure, an increase in stock options reduces R&D investment; by contrast, in late stages of CEO tenure, an increase in stock options fosters R&D spending. Overall, stock option granting reduces R&D intensity early in a CEO tenure, and increases it in later stages of CEO tenure.

Model 3 of Table 2 reports that the interaction of Early CEO tenure and CEO duality is positive and significant ($b = .008$; $p < .05$), providing support for Hypothesis 2. The coefficient indicates that the CEO duality increases R&D intensity in the early stages of CEO tenure, relative to late stages of CEO Tenure. The dummy variable of CEO duality is not standardized, and the coefficient reflects the change in R&D intensity associated with duality structures, relative to non-duality structures. The coefficients indicate that CEO duality reduces R&D intensity by .05% in late stages of CEO tenure, and increases it by .03% in early stages of CEO tenure. Fig. 2 depicts the

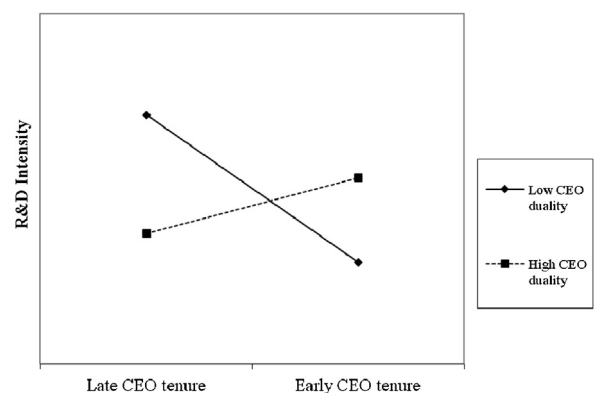


Fig. 2. Interaction effect: stages of CEO tenure X CEO duality.

Table 1
Variables, measurement and expected effect.

Variable	Measurement	Expected effect																
R&D intensity	R&D expenditures/total assets	Dep. var.																
CEO stock options	US Dollar value	– early in CEO tenure + late in CEO tenure																
CEO duality	Dummy (1 = combined CEO and Chair positions)	+ early in CEO tenure – late in CEO tenure																
Outsider ratio	Proportion of outside directors on total directors	– early in CEO tenure + late in CEO tenure																
Early CEO tenure	Dummy (1 = first 3 years in office)	±																
CEO tenure	Number of years in office	Inverted <i>U</i> -shaped																
Technological opportunities	Average industry R&D expenditures/total assets	+																
Industry Appropriability	Number of industry patents per industry/number of firms that patent in the industry	+																
Industry concentration	Herfindal index of competitors market shares in a given industry (2 digit SIC codes)	±																
Firm size	Log of total sales	±																
Firm age	Number of years since firm's foundation	–																
Ownership concentration	Perc. of shares held by blockholders	+																
Firm performance	Net income/equity	±																
CEO salary	US Dollar value	+																
CEO bonus	US Dollar value	+																
CEO scientific background	Dummy (1 = CEO with background in R&D)	+																
Board size	Number of directors	±																
Board committees	Dummy (1 = more than 3 committees)	+																
Industry dummies	Dummies (2 digit SIC codes)	±																
Year dummies	Dummies (1 per year)	±																
Descriptive statistics																		
Variables	Mean	Std. Dev	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1 R&D intensity	0.05	0.05																
2 Firm size	14.45	1.59	–0.22															
3 Industry Tech. Opport	0.00	1.00	0.16	0.04														
4 Industry cone.	0.25	0.20	–0.26	0.08	–0.13													
5 Industry Appopr.	0.00	1.00	–0.09	0.22	–0.05	0.05												
6 Ownership cone.	0.20	0.15	0.05	–0.26	–0.06	0.07	–0.08											
7 Firm age	48.03	39.15	–0.28	0.35	–0.03	0.13	0.11	–0.18										
8 Firm performance	0.14	0.42	–0.16	0.22	–0.01	0.06	0.02	–0.16	0.21									
9 CEO sci. Back.	0.31	0.46	0.22	0.04	0.02	–0.05	–0.04	–0.05	–0.01	–0.01								
10 CEO salary	757.72	381.93	–0.22	0.70	0.11	0.12	0.22	–0.20	0.36	0.23	–0.03							
11 CEO bonus	739.21	1432.58	–0.09	0.33	–0.02	0.06	0.14	–0.15	0.13	0.15	–0.01	0.29						
12 Board size	11.05	0.94	–0.22	0.50	0.09	0.13	0.15	–0.12	0.33	0.16	0.03	0.46	0.03					
13 Board committees	0.76	0.43	–0.12	0.22	0.03	0.09	0.01	0.02	0.14	0.14	–0.03	0.21	0.02	0.35				
14 CEO stock options	2840.77	3585.56	0.07	0.19	0.03	–0.03	0.02	–0.10	–0.01	0.02	0.04	0.05	0.13	–0.03	–0.08			
15 CEO duality	0.64	0.48	–0.08	0.18	0.06	0.07	0.05	0.03	0.13	0.07	–0.04	0.25	0.06	0.10	0.14	–0.04		
16 Outsider ratio	0.79	0.10	–0.10	0.25	–0.06	0.08	0.00	–0.03	0.16	0.09	0.01	0.23	0.10	0.13	0.31	0.02	0.21	
17 Early CEO tenure	0.48	0.45	–0.02	0.15	–0.01	–0.02	0.06	–0.01	0.10	0.00	–0.01	0.03	0.01	0.18	0.15	0.02	–0.23	0.04

Sig levels $|r| > .05$, $p < .05$, $r > |.08|$, $p < .01$.

Table 2

Regression results. Dependent variable: R&D intensity.

Variables	Model 1	Model 2	Model 3	Model 4
Interaction: outsider ratio X Early CEO tenure				–0.003 ⁺ (0.001)
Interaction: CEO duality Early CEO tenure			0.008 ⁺ (0.003)	0.011 ^{***} (0.003)
Interaction: CEO stock options X Early CEO tenure		–0.014 ⁺ (0.006)	–0.014 ⁺ (0.006)	–0.015 ⁺ (0.006)
Early CEO tenure (Dummy)	–0.001 (0.001)	–0.005 ⁺ (0.002)	–0.005 ⁺ (0.002)	–0.008 ^{***} (0.003)
Outsider ratio	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.001 ⁺ (0.001)
CEO duality	–0.002 (0.002)	–0.002 (0.002)	–0.005 ⁺ (0.002)	–0.006 ^{***} (0.001)
CEO stock options	0.000 (0.000)	0.010 ⁺ (0.005)	0.010 ⁺ (0.005)	0.010 ⁺ (0.005)
Board committees	–0.000 (0.001)	–0.000 (0.002)	–0.000 (0.002)	–0.000 (0.002)
Board size	–0.001 (0.001)	–0.001 (0.001)	–0.001 (0.001)	–0.001 (0.001)
CEO bonus	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
CEO salary	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 ⁺ (0.000)
CEO scientific background	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)
R&D intensity (lagged)	0.832 ^{***} (0.050)	0.837 ^{***} (0.049)	0.841 ^{***} (0.041)	0.835 ^{***} (0.042)
Firm performance	0.000 (0.001)	0.000 (0.001)	0.001 (0.002)	0.001 (0.002)
Firm age	–0.000 (0.000)	–0.000 ⁺ (0.000)	–0.000 ⁺ (0.000)	–0.000 ⁺ (0.000)
Ownership concentration	0.001 (0.003)	0.001 (0.003)	0.003 (0.004)	0.002 (0.003)
Industry Appropriability	0.000 (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Industry Concentration	–0.005 ⁺ (0.002)	–0.005 ⁺ (0.002)	–0.005 ⁺ (0.002)	–0.004 (0.003)
Industry Tech. Opport	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)	0.003 ⁺ (0.001)
Firm size	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	–0.000 (0.001)
Intercept	0.010 (0.009)	0.010 (0.009)	0.014 (0.008)	0.018 ⁺ (0.010)
Year dummies	Included	Included	Included	Included
Industry dummies	Included	Included	Included	Included
Hans en test	175.92	200.5	218.19	244.64
Wald chi-square tests	2869.267 ^{***}	3233.177 ^{***}	2943.530 ^{***}	3452.354 ^{***}

N = 2170. Sig levels: ⁺ $p < .10$; ⁺ $p < .05$; ⁺ $p < .01$; ^{***} $p < .001$.

effect: when a CEO is in Early stage of tenure, CEO duality increases R&D investment; the effect reverts in later stages of CEO tenure, so that CEO Duality decreases R&D investment. Thus, CEO duality increases R&D intensity early in a CEO tenure, and decreases it in later stages of CEO tenure.

Finally, Model 4 of Table 2 also reports that the interaction of Early CEO tenure and outsider ratio is negative and significant ($b = -.003$; $p < .05$), providing support for Hypothesis 3. The coefficient indicates that the outsider ratio reduces R&D intensity in early stages of CEO tenure, relative to late stages of CEO tenure. The variable is standardized: the standard deviation of outsider ratio reflects approximately – for an average board of 10 or 11 members – the additional presence of one outside directors. The coefficients indicate that such a change entails an increase in R&D intensity of 0.01% in late stages of CEO tenure, and a decrease of .02% in early stages of CEO tenure. Fig. 3 depicts the effect: in early stages of tenure, an increase in outsider ratio decreases R&D investment; however, in later stages of CEO tenure, an increase in outsider ratio increases R&D investment. Overall, outsider ratio reduces R&D intensity early in a CEO tenure, and increases it in later stages of CEO tenure.

Table 3 reports results for the quadratic, continuous specification. There is no evidence of a basic inverted-U shaped relation between CEO tenure and R&D investment: this results is consistent with previous research which has generally found a complex set

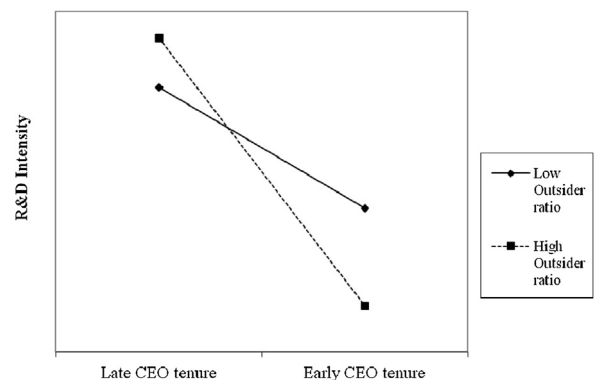
**Fig. 3.** Interaction effect: stages of CEO tenure X outsider ratio.

Table 3

Additional analyses. Dependent variable: R&D intensity.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Int.: CEO tenure sq.XOutsider ratio									0.000 (0.000)
Int.: CEO tenure X Outside ratio							0.000 (0.000)	0.001 (0.001)	0.001 (0.001)
Int.: CEO tenure sqX CEO Duality						–0.000 (0.000)			
Int.: CEO tenure X CEO duality				0.000 (0.001)	–0.002* (0.001)	–0.001* (0.001)			
Int.: CEO tenure sqXCEO stock opt.			–0.001** (0.000)						
Int.: CEO tenure X CEO stock opt.	0.001 (0.001)	0.000 (0.001)	0.003* (0.001)						
CEO tenure squared		–0.000 (0.000)	0.001* (0.000)		0.001* (0.000)	0.001* (0.001)		0.000 (0.000)	0.000 (0.000)
CEO tenure	–0.001 (0.001)	0.001 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.004* (0.002)	–0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)
Outsider ratio	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)
CEO duality	–0.001 (0.001)	–0.002 (0.001)	–0.001 (0.001)	–0.003* (0.001)	–0.001* (0.001)	–0.001* (0.001)	–0.003** (0.001)	–0.004** (0.001)	–0.004** (0.001)
CEO stock options	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Board committees	–0.001 (0.001)	–0.001 (0.001)	–0.001 (0.002)	–0.001 (0.001)	–0.000 (0.001)	–0.000 (0.002)	–0.001 (0.001)	–0.001 (0.001)	–0.001 (0.001)
Board size	–0.000 (0.001)	0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.001 (0.001)	–0.001 (0.001)
CEO Bonus	0.000* (0.000)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
CEO salary	–0.000 (0.000)	–0.000* (0.000)	–0.000* (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000* (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)
CEO scientific backgr.	0.001 (0.002)	0.002 (0.002)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.000 (0.002)	0.000 (0.002)
R&D intensity (lagged)	0.871*** (0.037)	0.878*** (0.037)	0.888*** (0.037)	0.815** (0.038)	0.823*** (0.036)	0.825 (0.034)	0.830** (0.038)	0.830** (0.038)	0.836*** (0.034)
Firm performance	0.001 (0.001)	0.001 (0.001)	0.001 (0.002)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Firm age	–0.000* (0.000)	–0.000* (0.000)	–0.000* (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)
Ownership cone.	–0.001 (0.003)	–0.001 (0.003)	–0.001 (0.003)	–0.001 (0.003)	–0.001 (0.003)	0.000 (0.003)	–0.001 (0.003)	–0.001 (0.003)	–0.001 (0.003)
Industry Appro pr.	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	–0.000 (0.000)	–0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Industry Cone.	–0.004* (0.002)	–0.003 (0.002)	–0.003 (0.002)	–0.006* (0.002)	–0.005* (0.003)	–0.006* (0.003)	–0.005* (0.002)	–0.005* (0.002)	–0.005* (0.002)
Industry Tech. Opport	0.003** (0.001)	0.003 (0.001)	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)	0.003 (0.001)	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)
Firm size	–0.000 (0.001)	–0.000 (0.001)	0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	–0.000 (0.001)	0.000 (0.001)	–0.000 (0.001)
Intercept	0.016* (0.008)	0.014* (0.009)	0.010 (0.008)	0.020* (0.009)	0.019* (0.008)	0.019* (0.009)	0.018* (0.009)	0.016* (0.009)	0.017* (0.009)
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
Hansen test	211.21	231.75	268.61	229.2	236.75	264.32	227.64	244.54	264.45
Wald chi-square tests	3243.395***	3271.547***	3840.162***	2132.965**	2327.175***	2457.297***	2201.570**	2238.795***	2292.235***

N = 2170. Sig. levels: *p < .10, **p < .05, ***p < .01; ****p < .001.

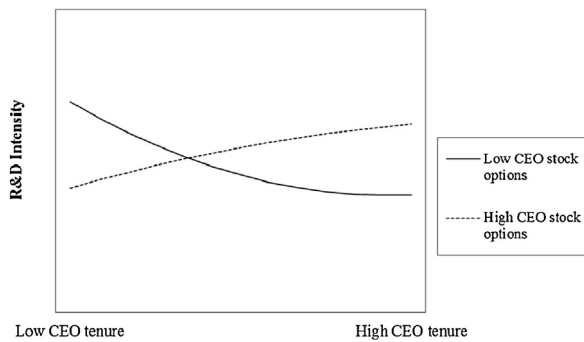


Fig. 4. Interaction effect: CEO tenure and stock options.

of relations, including no significant effect (Luo et al., 2014), linear effects (Barker and Mueller, 2002; Simsek, 2007), or quadratic effects (Wu et al., 2005; Miller and Shamsie, 2001). Specifically, Wu et al. (2005) find that the relation between CEO tenure and corporate invention (i.e., a measure of innovation output) becomes flat under average technological dynamism, which is consistent with our dataset constituted by firms from differing industries.

Model 3 reports that the quadratic effect is strongly significant for CEO stock options: both the interaction of CEO tenure and CEO stock options ($b = .003$; $p < .05$), and CEO tenure squared and CEO stock options ($b = -.001$; $p < .01$) are significant. Fig. 4 depicts the effect: as CEO tenure progresses, CEO stock options increase R&D intensity, yet, it does so at a decreasing marginal rate; by contrasts, CEOs with relatively limited amounts of stock options, reduce R&D investment to a greater extent, the longer their time in office. In addition, early in CEO tenure, larger stock options decrease R&D intensity, whereas the reverse occurs later in CEO tenure. Model 5 of Table 3 accounts for the effects of CEO duality: the interaction of CEO tenure and CEO duality is negative and significant ($b = -.002$; $p < .05$); Fig. 5 depicts the effect: as CEO tenure progresses, R&D investment decreases to a greater extent under duality. We did not find any significant effect for outsider ratio: while in the previous analysis we did find differences on average effect for outsider ratio, we do not find a more fine-grained relation of outsider ratio as CEO tenure progresses. One possible reason is that in United States outside directors are under close scrutiny by shareholders; hence, even under relatively reduced levels of outsider ratio, the board is active, mitigating the fine-grained effect of outsider ratio on decisions of R&D intensity, along a continuous measure of CEO tenure. This explanation is speculative, and more research is needed to inquire into this phenomenon.

We performed further tests, to ascertain whether there is a statistically significant difference in our predicted effects, when the CEO has vs. has not a scientific background, and when there is a large vs. small number of committees, but found no such an effect. Future research may further inquire into such phenomena.

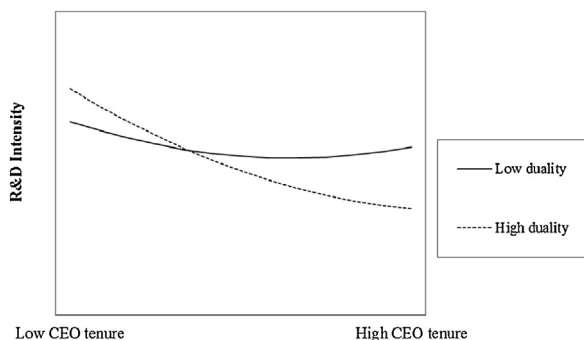


Fig. 5. Interaction effect: CEO tenure and CEO duality.

5.2. Further results

We have also explored whether the influence of governance devices changes across differing industries. To this aim, we have performed an additional test, to ascertain whether our hypothesized effects change in high-tech vs. low-tech industries. In high-tech industries, decisions on R&D investment are at the core of board discussion: the hypothesized effects of stock options and board independence will be magnified in these industries. In low-tech industries, governance devices do shape a CEO's decisions, but their effects will regard, to a greater extent, the general management, and will be less pronounced for R&D investment.

Specifically: high-tech industries are conceived as high-discretion contexts (Hambrick and Finkelstein, 1987), so that CEOs exert a major impact on decisions: the board acknowledges that “there can be a substantial performance difference that can emanate from a ‘good’ manager and a ‘bad’ manager” (Balkin et al., 2000, p. 1119). In addition, the board's assessment will be especially salient for new CEOs: in high-tech industries, “CEOs face a greater probability of termination and a higher likelihood of a negative

Table 4

Subgroup analysis: high tech vs. Low tech industries. Dep var.: R&D intensity.

Variables	High tech	Low tech
Interaction: outsider ratio X Early CEO tenure	−0.005* (0.003)	−0.002 (0.001)
Interaction: CEO duality Early CEO tenure	0.005 (0.008)	0.005 (0.005)
Interaction: CEO stock options X Early CEO tenure	0.000 (0.003)	0.002 (0.002)
Early CEO tenure (Dummy)	0.001 (0.002)	−0.001 (0.004)
Outsider ratio	0.001 (0.002)	0.001 (0.001)
CEO duality	−0.006 (0.005)	−0.006* (0.004)
CEO stock options	−0.000 (0.001)	−0.001 (0.001)
Board committees	0.002 (0.002)	−0.004* (0.003)
Board size	−0.004* (0.002)	0.001 (0.001)
CEO bonus	0.000 (0.000)	0.000 (0.000)
CEO salary	0.000 (0.000)	0.000 (0.000)
CEO scientific background	0.003* (0.002)	0.002* (0.002)
R&D intensity (lagged)	0.857*** (0.077)	0.847*** (0.058)
Firm performance	0.012* (0.007)	−0.000 (0.001)
Firm age	−0.000 (0.000)	−0.000* (0.000)
Ownership concentration	0.011* (0.007)	−0.001 (0.003)
Industry Appropriability	0.000 (0.001)	−0.000 (0.000)
Industry concentration	−0.009* (0.006)	−0.001 (0.002)
Industry Tech. Opport	0.002 (0.001)	0.001 (0.001)
Firm size	0.000 (0.001)	−0.001 (0.001)
Intercept	0.005 (0.015)	0.024 (0.020)
Year dummies	Included	Included
Industry dummies	Included	Included
Hansen stat	99.66	130.10
Wald chi-square stat	1215.30***	1906.95***

N = 2170. Sig levels: * $p < .10$; * $p < .05$, ** $p < .01$; *** $p < .001$.

High tech industries: 132 firms; low tech industries: 178 firms.

reputation” in case of failure (Balkin et al., 2000, p. 1119). As a result: it can be expected that, early in a CEO tenure, stock options and board independence will reduce R&D investment to a greater extent, in high-tech industries. In later stages of CEO tenure, the difference should attenuate, because CEO's accumulated power will prevail, relative to external conditions.

To test for this possibility, we split our sample in two subgroups, high-tech and low-tech, according to the classification by Hall and Vopel (1997) for United States, resulting in 132 firms in high-tech industries, and 178 in low-tech industries. Table 4 reports the results: it shows that, as regards CEO stock options and CEO duality, there is no significant difference in high-tech vs. low-industries. Yet, we do find that the hypothesized effect of outsider ratio holds for high-tech industries ($b = -.005$; $p < .05$), but it is non-significant in low-tech industries; the coefficient is larger in the high-tech subsample, and a chow test ($p < .10$) confirms this difference. The larger, negative coefficient of the interaction, relative to low-tech industries, indicates that, early in CEO tenure, outsider ratio decreases R&D investment especially in high-tech contexts. In later stages of tenure, the effect is less pronounced.

6. Discussion and conclusions

Research on innovation points to corporate governance as one key determinant of R&D investment: CEOs' interests are not aligned with those of shareholders, so that incentive compensation and strong controls are needed to alter CEOs' motives and boost R&D spending (Ahuja et al., 2008). But empirical research has yielded inconclusive results (Kor, 2006; Driver and Guedes, 2012). Shedding light on how R&D investment can be enhanced by governance devices is important, to prevent rules from inadvertently impairing R&D spending.

6.1. The impact of governance devices on R&D along the CEO tenure

This study clarifies that *governance's impact on innovation effort is time-dependent, and is contingent upon CEO tenure*. These effects hold for both stock options and board independence, and, for the most part, also considering a continuous measure of CEO tenure, as shown in our additional analyses.

Early in a CEO's tenure, large amounts of stock options reduce R&D investment. Newly appointed CEOs face significant challenges under conditions of limited competence. In such circumstances, how they get paid may influence their proclivity to foster R&D, but the effect of stock options differ from what generally envisaged by previous research: large stock options aggravate CEOs' perceived risk and alter their problem framing, leading to reduced R&D investment. This finding warns boards of directors, shareholders, and consultants that great care needs to be taken with stock option awards to newly appointed CEOs: stock options are often regarded as a mechanism to attract new CEOs, because they offer major opportunities for wealth's increases in the future, but this function is not without costs: our study outlines that, while engaging in negotiations to hire a new CEO, boards of directors should be cognizant that large stock option awards, early in a CEO tenure, may dampen a CEO proclivity to invest in R&D.

This study also indicates that, in later stages of CEO tenure, the effect of stock options on R&D investment reverses: large stock options increase R&D investment. This finding has three key implications for stock options and R&D investment. First, our study suggests that R&D investment can be fostered, via stock option awards, not so much because of options' inherent, technical features (e.g., their largest potential for wealth increases), as because stock options counteract a greater tendency of CEOs to minimize R&D investment as CEO tenure progresses: hence, the

incentive properties of stock options, as a means to boost R&D spending, need to be understood *in reference to CEO psychological attitudes and inclinations, as they evolve along CEO tenure*. Second, while most research on innovation relies on the dominant agency view, and its idea that financial incentives align CEOs' interests with those of shareholders, an emerging stream of research has developed an opposite view, to suggest that CEOs seek to protect their real (or perceived) wealth, so that large stock options increase risk bearing and reduce R&D investment (Wiseman and Gomez-Mejia, 1998). Our study helps reconcile these conflicting theses, pointing to CEO tenure as a key contextual factor to catch stock option influence on R&D investment.

Third, our finding gives rise to an interesting research question for future inquiries, and a caveat for practitioners. In early CEO tenure, boards may seek to minimize the negatives of stock options by promising option awards in future times, after an early transition periods. However, the behavioral agency view, and its idea of instant endowment (Wiseman and Gomez-Mejia, 1998), suggests that, if a new CEO expects to be given stock options in the future, he/she will adjust behaviors accordingly in the present: a promise of future stock options grants may thus be as detrimental for R&D investment as actual awards in current times. This idea is in need of further investigation, and such an inquiry may lead to reconsider the policy of stock option granting along CEO tenure.

The effects of board independence on R&D investment are also time dependent. As our study shows, early in a CEO tenure the agency problem of limited competence is most salient, and the effects of board dependence/independence are significantly altered: contrary to standard agency predictions, early in CEO tenure, nonduality and outsider ratio may reduce innovation effort. Driver and Guedes (2012) examine board independence on a sample of companies in UK, and found that strong controls dampen, rather than fostering, R&D investment; they conclude for “the importance of managerial security and autonomy, if risky investments are to be sustained” (2012, p. 1575). Our study is complementary to their finding, and outlines under which conditions such policy should be ensured: for R&D spending to be increased via governance devices, a CEO needs autonomy and discretion especially in the early years of tenure; later in CEO tenure, board independence becomes beneficial.

One question for future research, with implications for R&D investment, is whether and how a board can shift from dependence to stronger independence as CEO tenure progresses, provided that CEO power increases with tenure and may possibly hinder an increase in board independence over time. If a shift toward greater board independence is to occur, it is more likely to happen immediately after a CEO has proven his/her leadership in office; Miller and Shamsie (2001) and Hambrick and Fukutomi (1991) have outlined how, after the first three years in office, executives overcome their initial uncertainty on the job and reinforce their position. Performance deficits might offer an alternative opportunity for a shift in governance design: poor performance weakens CEOs, so that the board might negotiate a change in board design as an alternative to dismissal, and future studies might inquire into these dynamics. From the practical point of view, situations will vary across firms, and the assessment of the specific conditions of CEO reappointment and power balance ultimately relies on the board and shareholders, who need to design the best governance devices to boost effective innovation.

One final contribution regards the evolving trend of R&D spending along a CEO tenure. Previous research has emphasized that R&D investment is affected *directly* by change over time in the CEO's psychological attitudes: Barker and Mueller (2002) found that R&D investment decreases monotonically as CEO tenure progresses; Wu et al. (2005) found an inverted U-shaped relationship between tenure and patent filing, which first increases and then

decreases as tenure lengthens. However, Luo et al. (2014) have recently asserted that “the impact of CEO tenure is a complex phenomenon that goes beyond the simple direct effects” (pp. 492–493) and that “to get a holistic view of the causal linkages between CEO tenure and firm performance, it is important to explore the underlying mechanisms that explain how CEO tenure matters” (p. 493; see also Simsek, 2007). We show that the relation between R&D investment and CEO tenure is not direct and uniform for all CEOs, but changes depending on governance mechanisms, such as stock options and board independence.

6.2. Further implications for research on innovation and governance devices

Our findings challenge Hendry's conclusions that “the effects of incompetence may be largely indistinguishable from the effects of opportunism” (2002, p. 107), and that the simplified assumption of *homo economicus* “can be justified on heuristic grounds” (p. 98). We have seen that, in specific management situations, stock options and board independence may harm shareholders' interests and actually create an agency problem, rather than solving it. Hence we contend that standard agency theory's assumption of *homo economicus* may not have universal validity, and its predictions do not always hold in a real world. Agency scholars should pay attention to the nature of principal-agent relations, and to whether the circumstances are mostly conducive of an agency problem of limited competence, rather than managerial opportunism.

Scholars in innovation rely on standard agency theory, and constantly point to the inflexible conflict of interests in principal-agent relations, to predict that stock options and board independence will be effective regardless of the specific circumstances under which principal-agent relation comes into play. Our study challenges such a universal view: this study shows that *principal-agent relations are context-dependent, and agency theory should be applied taking into consideration the specific circumstances in which principal and agent operate*. This conclusion has implications for future research, as argued below.

Board dependence can serve shareholders' interests, and this is totally consistent with agency theory and the minimization of agency costs. In agency relations primarily characterized by limited competence, granting greater discretion to a CEO and ensuring pay security (i.e., board dependence and a decoupling of CEO pay to firm performance) can be conceived as a cost that the principal may be willing to incur in order to avoid the cost of incompetence (a reduced proclivity toward risk and innovation). In principal-agent relations affected by problems of limited competence, it will pay the principal to design governance mechanisms that enhance agent discretion and favor the freedom to experiment and learn.

Moreover, this study advances cost-benefit analysis of independent board structures. Beatty and Zajac (1994, p. 313) note that agency scholars often analyze “the benefits, but not necessarily the costs of the monitoring technology observed in modern corporations.” This study helps clarify the implicit costs associated with independent board structures: they ensure strong control against managerial opportunism, but they may also engender costs that are particularly salient under conditions of limited competence and high ambiguity in the principal-agent relation.

6.3. Implications for policy makers, practitioners, and stockholders

The vast body of regulations, law, governance codes, and standards of good corporate governance has been widely influenced by the agency-theory's claim that managers must be controlled, for them to boost R&D investment. In contrast, some recent studies prescribe a general relaxation of control. These universalistic

prescriptions state “best practices” for a plethora of organizations facing differing circumstances. In fact, our study shows that, in the early stage of CEO tenure, board dependence and a decoupling of CEO pay from financial performance boost R&D spending; this result is significant, considering “the disproportionate number of CEOs, whose tenure last 3 years or less” (Hambrick and Fukutomi, 1991). In order to improve R&D investment, in the future, policy makers and regulators should take a more detailed, fine-grained approach and tailor rules for firms under specific circumstances, recommending incentives/control in later stages of CEO tenure, when the problem of managerial opportunism becomes most relevant. What is needed is a richer set of standards for good governance, suggesting control/autonomy for differing firms depending on circumstances.

6.4. Boundary conditions and limitations

This study seeks only to discover general tendencies: its validity for the single corporation will depend on the situation facing each executive. In addition, tenure is an indirect measure of motivation; future research would certainly benefit from a direct measure of CEO psychological attitudes along the CEO tenure cycle. In the words of Simsek (2007), “we built our theory on the fundamental assumption that CEO tenure adequately captures aspects of executive attention and behavior . . . collecting data on such constructs might prove difficult” (Simsek, 2007, p. 661). Moreover, “while fine-grained psychological constructs may better capture underlying beliefs and preferences, observable CEO characteristics, such as tenure, might have superior value, because decision makers must make judgments about competitors, based on what can be observed, i.e., their visible characteristics” (Simsek, 2007, p. 662).

In accordance with previous research (Pfeffer, 1983; Dalton et al., 2008), we rely on board characteristics to assess how different configurations of board structure impact R&D investment, but more research is needed to clarify the determinants of R&D: for example, inquiring into the internal composition and functioning of board committees may shed further light on R&D investment (Upadhyay et al., 2014; Lejarraga and Martinez-Ros, 2014). A recent stream of research emphasizes the role of internal dynamics, behaviors and decision making styles (Forbes and Milliken, 1999; Zona and Zattoni, 2007): future studies may inquire into how R&D investment is shaped by these processes, along the CEO tenure cycle. In addition, future studies may also inquire further into the role of CEO scientific background: our exploratory analysis has not provided evidence in this regard, but further consideration of this issue is needed.

Another limitation is that our findings pertain to public companies in United States, but governance systems vary across countries: stock options, CEO duality, and outsider ratio may be more or less prominent in differing contexts. For example, countries such as Germany, Netherlands, Japan have very different governance structures. There is a need of cross-national studies on corporate governance; future studies may draw from the mechanisms and theory of this study to ascertain whether the effects change across countries. In addition, an extended typology of governance devices, such as shareholders' rights, cross-listing, control pyramid (Lhuillery, 2011), may affect CEO's motives to invest in R&D: future research may inquire into such phenomena, and ascertain whether their effect differs in early vs. later stages of CEO tenure.

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